BIOLOGY COLLOQUIUM SERIES | FALL 2026



Unraveling Marine Biodiversity Across Scales Through Functional Traits

Mon, November 2nd
12:00 – 1:30 PM
RB 130

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Understanding how marine communities respond to environmental change requires linking species characteristics to the processes that organize ecosystems. In this seminar, I will discuss how functional traits provide a powerful framework for connecting observations across multiple scales, from individual organisms and local assemblages to oceanic islands and broad biogeographic regions. Using reef fishes as a central model, I will draw on field surveys, long-term monitoring, ecological interactions, and large comparative datasets from the Atlantic and Eastern Pacific. These studies illustrate how life-history strategies, ecological roles, geographic distributions, and environmental conditions shape community composition and ecosystem functioning. I will explore why locations with similar levels of species richness may differ substantially in ecological structure, how warming and other anthropogenic pressures reorganize communities, and how isolation and habitat gradients influence the assembly of island faunas. The seminar will also address the value of integrating functional information with taxonomic and phylogenetic diversity, as well as the methodological challenges associated with comparing data collected at different spatial and temporal resolutions. Collectively, this work demonstrates how trait-based ecology can move beyond species inventories to identify mechanisms of ecological change, strengthen biodiversity assessments, and inform monitoring and conservation strategies for marine ecosystems facing accelerating environmental and anthropogenic pressure